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LIST OF ABBREVIATIONS

AI – Artificial Intelligence

API – Application Programming Interface

AWS – Amazon Web Services

CO – Course Outcome

CSV – Comma-Separated Values

ERP – Enterprise Resource Planning

ETL – Extract, Transform, Load

IAM – Identity and Access Management

KPI – Key Performance Indicator

NLP – Natural Language Processing

S3 – Simple Storage Service

AIEPS – AI Based Examination Proctoring System

UI – User Interface

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CHAPTER 1

INTRODUCTION

1.1 DATA ENGINEERING

1.1.1 What is data engineering

Data engineering is the practice of designing, building, and maintaining systems and infrastructure that collect, store, and process data for analytical and operational use in scalable manner that supports both real-time processing and analytical workflows. Cloud-based services such as Amazon RDS, Amazon S3, and Amazon Athena play a crucial role in enabling efficient data storage, querying, and analysis, as highlighted in AWS documentation and modern data engineering practices [5][6][7].

1.1.2 Types of data models

Data storage in modern applications can be broadly classified into relational databases, which use structured schemas and SQL for querying, and non-relational databases, which offer flexible, document-based or key-value storage [1]. Relational databases such as PostgreSQL and MySQL are preferred for applications that require strict data integrity and complex relationships, making them ideal for managing user accounts, vault entries, and authentication records in a password manager [2][4].

1.1.3 Data pipeline models

Data pipelines define how data flows from its source to its destination through a series of processing steps. Batch processing handles large volumes of data at scheduled intervals, while stream processing enables continuous, real-time data handling. AWS Glue supports ETL (Extract, Transform, Load) operations, and Amazon EC2 provides the compute environment needed to run custom database configurations and preprocessing scripts.

In the context of this project, a hybrid approach is used where credential data is processed in real time during user interactions, while machine learning model inputs such as login patterns and vault statistics are prepared through scheduled preprocessing pipelines [3].

1.1.4 AWS Services

Amazon Web Services provides a range of tools that support data engineering workflows. Amazon RDS offers managed relational database hosting, while Amazon S3 serves as scalable object storage for backups and model artifacts [8][9]. AWS Glue supports ETL (Extract, Transform, Load) operations, and Amazon EC2 provides the compute environment needed to run custom database configurations and preprocessing scripts. In this project, EC2 Ubuntu is used to host the SQLAlchemy-managed database along with all supporting data scripts.

1.2 AI ENGINEERING

1.2.1 What is AI Engineering?

AI engineering is the discipline of designing, developing, and deploying machine learning and artificial intelligence models into production systems in a reliable and scalable manner. It combines principles from software engineering, data science, and operations to ensure that AI models function correctly in real-world environments. In a password manager, AI engineering enables the system to go beyond static rule-based security by learning from user behavior and data patterns to provide dynamic, intelligent protection.

1.2.2 Types of Machine Learning Models

Machine learning models can be categorized into supervised learning models, which learn from labeled data to make predictions, unsupervised learning models, which identify patterns without predefined labels, and semi-supervised models that combine both. In this project, supervised regression models are used for password strength prediction, unsupervised Isolation Forest models detect login anomalies, TF-IDF based models handle URL categorization and phishing detection, and clustering algorithms identify similar or reused passwords within the vault.

1.2.3 Model Deployment Approaches

AI models can be deployed in several ways depending on latency and resource requirements. Batch inference processes predictions periodically on accumulated data, while real- time inference serves predictions instantly during user interactions. Edge deployment runs models locally on the device, whereas cloud-based deployment hosts models on remote servers.

This project uses real-time cloud-based inference where pre-trained scikit-learn models are loaded into the Flask application on EC2 and queried during login events and vault operations.

1.2.4 AWS Services for AI Engineering

AWS offers several services that support AI and ML workflows. Amazon SageMaker provides an end-to-end platform for training, tuning, and deploying models. Amazon Comprehend offers natural language processing capabilities, and AWS Lambda enables serverless model inference. In this project, models are trained offline using scikit-learn and deployed directly on EC2, leveraging the instance's compute capacity for low-latency real-time inference without additional managed ML services.

1.3 CLOUD ENGINEERING

1.3.1 What is Cloud Engineering?

Cloud engineering is the application of engineering principles to design, build, and manage cloud-based infrastructure and services. It involves provisioning virtual servers, configuring networks, managing storage, and deploying applications in a way that maximizes reliability, scalability, and security. For a password manager that handles sensitive user data, cloud engineering ensures that the hosting environment is properly hardened, consistently available, and capable of supporting encrypted communications and real-time operations.

1.3.2 Types of Cloud Computing

Cloud computing is broadly classified into three service models. Infrastructure as a Service (IaaS) provides virtualized computing resources such as servers and storage over the internet. Platform as a Service (PaaS) offers a managed environment for application development and deployment. Software as a Service (SaaS) delivers fully hosted applications to end users. This project primarily utilizes IaaS through AWS EC2, giving full control over the operating system, runtime environment, and application configuration.

1.3.3 Cloud Deployment Models

Cloud deployments can follow public, private, hybrid, or multi-cloud models. A public cloud deploys resources on shared infrastructure managed by a provider like AWS. A private cloud runs on dedicated infrastructure for a single organization. A hybrid cloud combines both, while a multi-cloud strategy uses services from multiple providers. This project uses a public cloud deployment on AWS, where the application is hosted on a shared EC2 instance configured with strict security controls to simulate a production-grade private environment.

1.3.4 AWS Services for Cloud Engineering

AWS provides a comprehensive set of services for cloud infrastructure management. Amazon EC2 offers scalable virtual machines for hosting applications. Amazon VPC enables isolated network environments with custom routing and access rules. Elastic Load Balancing distributes traffic across instances for high availability. AWS IAM manages identity and access control, and Amazon Route 53 handles DNS management. In this project, EC2 is the primary service used alongside security groups, Nginx, Gunicorn, and UFW firewall for a secure and efficient deployment.

1.4 SECURITY ENGINEERING

1.4.1 What is Security Engineering?

Security engineering is the field of applying systematic engineering principles to build systems that are resilient against unauthorized access, data breaches, and malicious attacks. It covers a wide range of practices including cryptography, authentication, access control, and secure system design. In the context of a password manager, security engineering is the most critical discipline as the entire purpose of the application is to safeguard sensitive user credentials against both current and emerging threats.

1.4.2 Types of Cryptographic Techniques

Cryptographic techniques can be divided into symmetric encryption, where the same key is used for both encryption and decryption, and asymmetric encryption, which uses a public-private key pair. Hashing is a one-way transformation used to store passwords securely without retaining the original value. Post-quantum cryptography is an emerging category that designs algorithms resistant to attacks by quantum computers. This project uses AES-256-EAX for symmetric encryption, Argon2 and scrypt for password hashing, and Kyber KEM for post-quantum key encapsulation.

1.4.3 Authentication Models

Authentication models define how a system verifies the identity of its users. Single-factor authentication relies solely on a password, while multi-factor authentication (MFA) requires additional verification such as a one-time code or biometric input. Passwordless authentication uses hardware tokens or device-based verification. This project implements a layered authentication model combining traditional password login with TOTP-based two-factor authentication and WebAuthn, which enables hardware-backed passwordless login for enhanced security.

1.4.4 AWS Services for Security Engineering

AWS provides a robust set of security services to protect cloud-hosted applications. AWS IAM controls access to resources through role-based policies. AWS Shield and AWS WAF protect against DDoS attacks and web exploits. AWS KMS manages cryptographic keys securely, and Amazon GuardDuty provides intelligent threat detection. AWS Security Groups act as virtual firewalls controlling instance-level traffic. In this project, EC2 security groups, UFW firewall, and HTTPS configurations form the primary security perimeter, with application-level security handled through custom cryptographic implementations using liboqs and standard libraries.

CHAPTER 2

PROBLEM STATEMENT

Data engineering forms the backbone of the Campus Wi-Fi Usage Monitoring and Anomaly Detection System by handling the collection, transformation, and storage of campus network usage data. Using Amazon API Gateway and AWS Lambda, the system can receive real-time Wi-Fi metrics such as device count, bandwidth consumption, access point load, session activity, and building-wise traffic. This data is then stored in Amazon S3 for scalable storage and organized into structured datasets that can be queried efficiently using Amazon Athena for reports and analytics. [1][2][3][4]

Cloud engineering ensures scalability, reliability, and continuous availability of the entire monitoring platform. With AWS Lambda and API Gateway, the backend automatically scales based on incoming network requests without managing physical servers. Amazon S3 provides highly durable storage for logs and historical records, while Amazon CloudWatch monitors application health, errors, and performance metrics in real time. This architecture ensures that multiple campus buildings and thousands of connected users can be monitored simultaneously without bottlenecks. [1][2][3][5]

AI engineering drives the intelligence of the system. Using threshold-based classification and anomaly detection logic, the platform can identify overloaded access points, unusual spikes in bandwidth usage, and abnormal connection behavior. Historical usage data can also be analyzed to estimate peak hours for each building. A smart assistant can answer queries such as Which building has highest traffic? or Which access point is overloaded now? and provide actionable insights for network administrators. [8][9][10][14]

Security engineering ensures safe handling of network analytics data and trusted access to cloud resources. With AWS Identity and Access Management (IAM), strict role-based permissions can be enforced for administrators and backend services. Amazon S3 encryption protects stored Wi-Fi logs and reports, while secure HTTPS communication protects data in transit. Amazon CloudWatch logging supports monitoring and auditing of system operations to improve accountability and reliability. [1][5][6].

OBJECTIVES

- Real-Time Wi-Fi Usage Data Collection and Monitoring
- Cloud Infrastructure Deployment and Hardening
- AI-Powered Anomaly Detection and Traffic Analysis
- Intelligent Query Assistant for Network Insights
- Secure Data Storage, IAM Access Control, and Monitoring
- Efficient Resource Management for Better Campus Connectivity
- Post-Quantum Cryptographic Security Implementation

CHAPTER 3

SYSTEM ARCHITECTURE

3.1 INTRODUCTION TO SYSTEM ARCHITECTURE

The proposed Campus Wi-Fi Usage Monitoring and Anomaly Detection System is designed as a modular, cloud-native architecture that integrates data engineering, cloud services, artificial intelligence, and security engineering within the AWS ecosystem. The goal of the system is to create a scalable, cost-effective, and intelligent platform that helps monitor campus network performance, detect unusual usage patterns, and improve connectivity management with minimal operational overhead. By leveraging AWS services, the system enables real-time data collection, analytics, automated anomaly detection, and secure handling of network usage data.

At the foundation, the data engineering layer manages the ingestion and transformation of Wi-Fi usage metrics such as connected device count, bandwidth usage, access point load, building-wise traffic, and timestamps. Services such as Amazon API Gateway and AWS Lambda support real-time data collection pipelines, while Amazon S3 provides centralized storage for historical records. Amazon Athena is used for structured SQL-based analytics over stored data. This ensures that raw network data is consistently processed, organized, and made available for anomaly detection and intelligent reporting.

To achieve this, the system leverages AWS services in a layered architecture. Incoming network data is collected through APIs, processed in Lambda, stored in S3, analyzed in Athena, and monitored through CloudWatch. The frontend dashboard built using React displays insights to administrators in an easy-to-understand format

Key Points :

- **Data Engineering**: Real-time ingestion through API Gateway, processing via Lambda, storage in S3, analytics through Athena
- **Cloud Engineering**: Real-time ingestion through API Gateway, processing via Lambda, storage in S3, analytics through Athena.
- **AI Engineering**: Threshold-based anomaly detection, peak usage prediction, smart assistant for network queries..
- **Security Engineering**: Threshold-based anomaly detection, peak usage prediction.

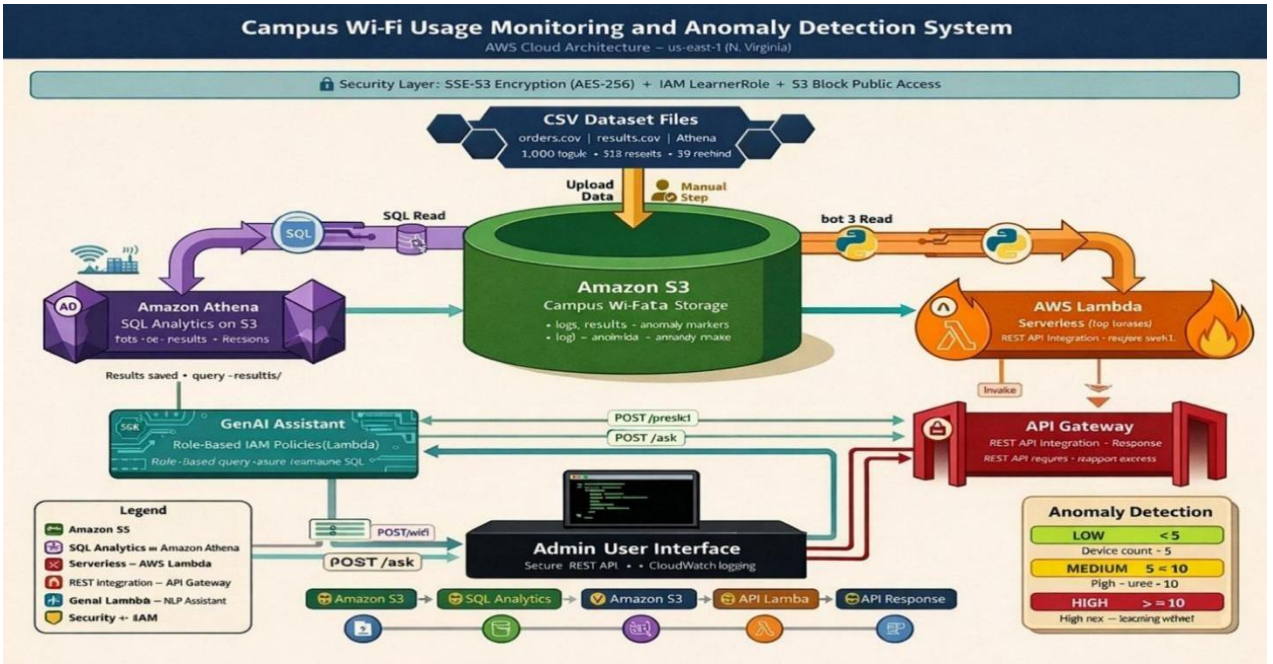


Figure 3.1 System Architecture

Fig. 3.1 depicts the system architecture of the Campus Wi-Fi Usage Monitoring and Anomaly Detection System, where Wi-Fi usage data is collected, stored in Amazon S3, and processed using AWS services.

3.2 CLIENT LAYER AND NETWORK SECURITY

The client layer represents the user interface through which administrators interact with the system. It consists of a web dashboard developed using React. Users can view live network statistics, alerts, reports, and ask queries about network conditions.

The client layer performs the following operations:

- Displays total connected devices.
- Shows overloaded access points.
- Displays building-wise traffic reports.
- Shows anomaly alerts.
- Provides smart query input for users.

To maintain secure access, the following mechanisms are implemented:

- Secure login for authorized users only.
- HTTPS communication for encrypted data transfer.
- Controlled API access using IAM permissions.
- Role-based dashboard access.

Although frontend controls improve security, backend validation is also used to ensure safe access and trusted communication.

Although client-side security enhances monitoring, it is complemented by backend verification to ensure reliability. Network security plays a crucial role in protecting data transmission between the client layer and backend services. Since the system handles sensitive information such as video streams, student data, and exam logs, secure communication is essential. All communication between the client and backend is performed over HTTPS, ensuring encryption of data in transit. This prevents interception or tampering of sensitive information such as event logs and video streams

3.3 APPLICATION LAYER

The application layer is the main processing unit of the Campus Wi-Fi Monitoring System. It handles business logic, processes incoming requests, performs analytics, and connects the frontend with AWS backend services.

The key functions of the application layer are:

- Collecting Wi-Fi metrics from data sources.
- Processing incoming usage records.
- Detecting overloaded access points.
- Identifying unusual traffic spikes.
- Running SQL analytics queries.
- Returning results to the frontend dashboard.
- **/collect-data** – Receives Wi-Fi metrics.
- **/analyze** – Runs anomaly detection and analytics.
- **/dashboard** – Sends summary data to frontend.
- **/ask-query** – Answers user questions.

This layer acts as a bridge between real-time monitoring and historical analytics. It ensures smooth communication and reliable system performance.

3.4 DATA LAYER AND BENEFITS

The data layer is responsible for storing and managing all Wi-Fi usage data generated by the campus network. It stores both structured and semi-structured data using cloud storage services.Amazon S3 is used to store usage logs, historical records, and analytics outputs. Amazon Athena is used to run SQL queries directly on stored data. This removes the need for a separate database server.

The data layer stores the following categories:

- Usage Logs – Device count, bandwidth, AP load, timestamps.
- Analytics Results – Reports, summaries, anomaly outputs.
- System Logs – API requests, Lambda logs, alerts.

Benefits of Data Layer

- Scalable storage for large volumes of logs.
- Fast analytics using SQL queries.
- Historical trend analysis.
- Reliable and secure cloud storage.
- Better reporting and decision making.
- Supports anomaly detection models.

Data Integrity

The system maintains organized folder structures, secure access permissions, encrypted storage, and reliable backup mechanisms to ensure data safety and consistency.

CHAPTER 4

MODULE 1 : DATA ENGINEERING

4.1 AIM

To design and implement a secure, scalable, and efficient cloud-based data engineering pipeline using AWS services for the Campus Wi-Fi Usage Monitoring and Anomaly Detection System. The module aims to collect, process, store, and analyze campus Wi-Fi usage data such as device count, bandwidth usage, access point load, and session logs while ensuring data integrity, optimized query performance, and seamless integration with analytics and machine learning modules.

4.2 PROCEDURE FOR DATA ENGINEERING

Step 1: Launch and activate the AWS Academy Learner Lab environment and access the AWS Management Console.

Step 2: Create an Amazon S3 bucket to store Wi-Fi usage logs, historical datasets, and analytics outputs with proper folder structure.

Step 3: Configure AWS Lambda functions to receive incoming Wi-Fi metrics such as device count, bandwidth usage, access point load, and timestamps.

Step 4: Create REST API endpoints using Amazon API Gateway to securely ingest real-time Wi-Fi data from frontend or simulated sources.

Step 5: Design structured datasets in CSV or JSON format for storing building-wise network usage records inside Amazon S3.

Step 6: Configure AWS Lambda to clean, transform, and organize incoming data before storing it in Amazon S3.

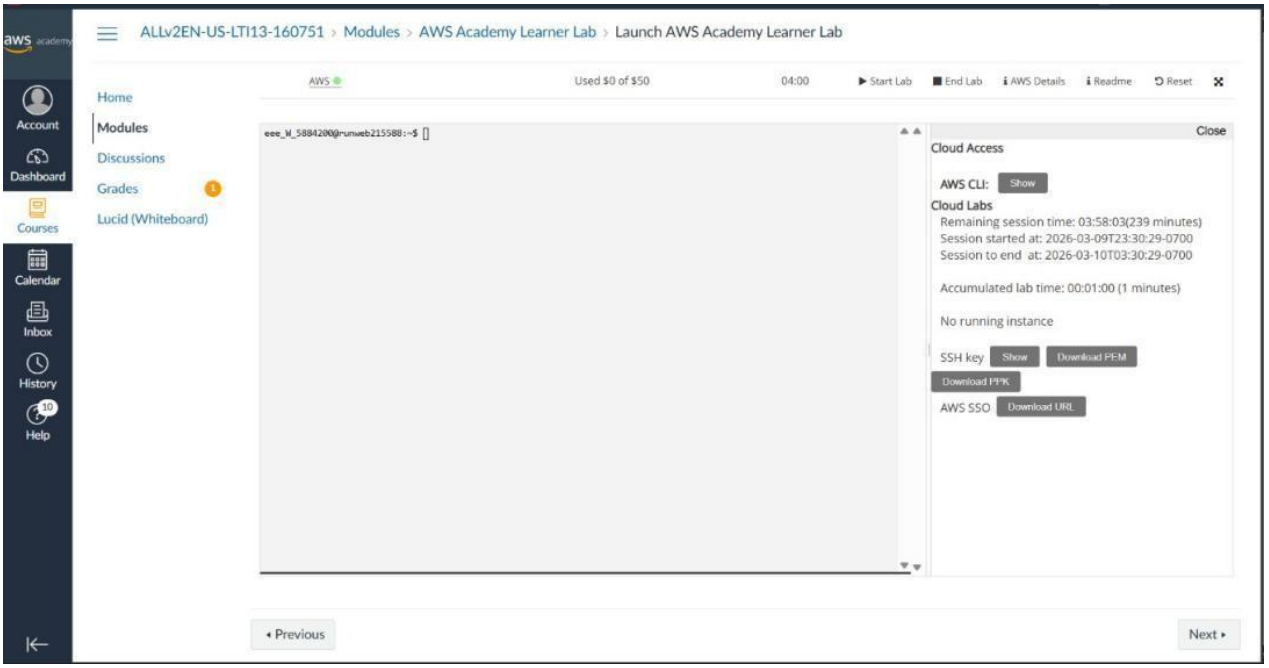
Step 7: Configure AWS Lambda to clean, transform, and organize incoming data before storing it in Amazon S3.

Step 8: Optimize query performance by partitioning data based on date, building, or access point location for faster analytics..

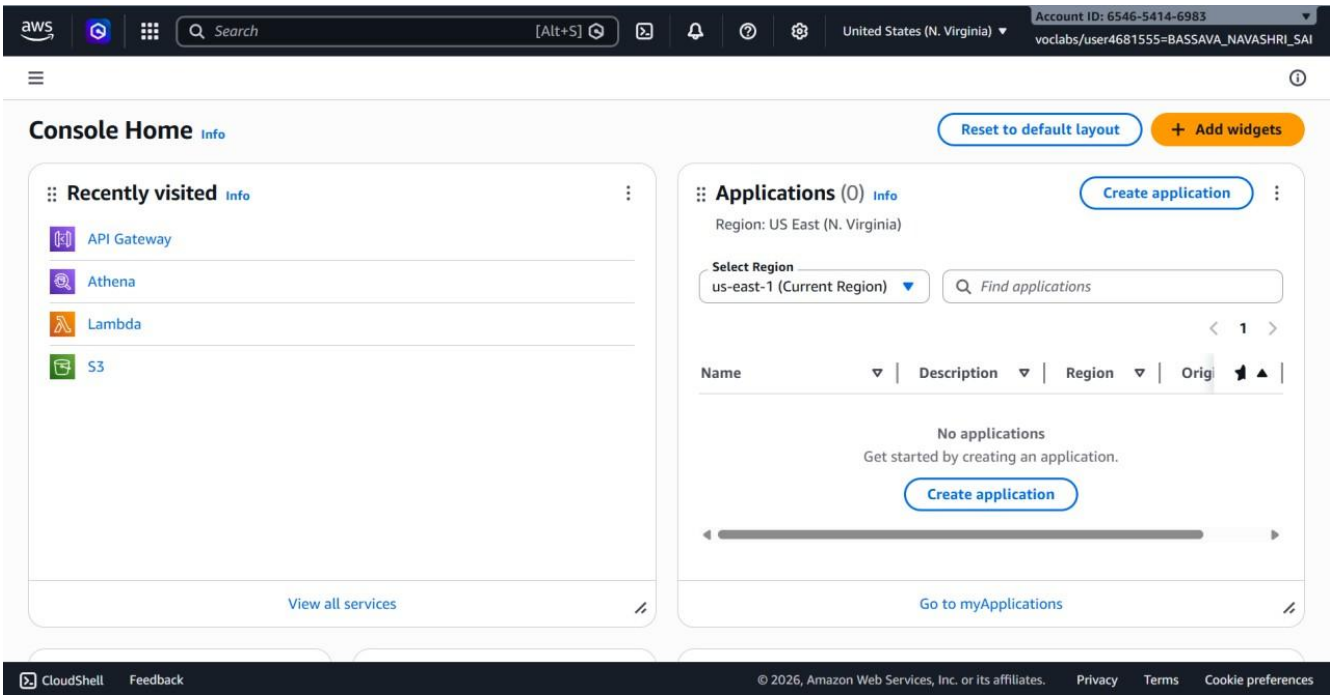
Step 9: Validate data consistency by querying stored records in Athena and verifying metrics such as overloaded access points, peak traffic hours, and bandwidth trends.

4.3 SCREENSHOTS FOR DATA ENGINEERING

The following screenshots demonstrate the key activities performed in Data Engineering:



Screenshot 4.1 : AWS Academy Learners Lab Status Active



Screenshot 4.2 : AWS Console Home - Access to Core Services Used in the Project

4.4 ACTIVITY DIAGRAM

Activity Diagram for Data Engineering Module

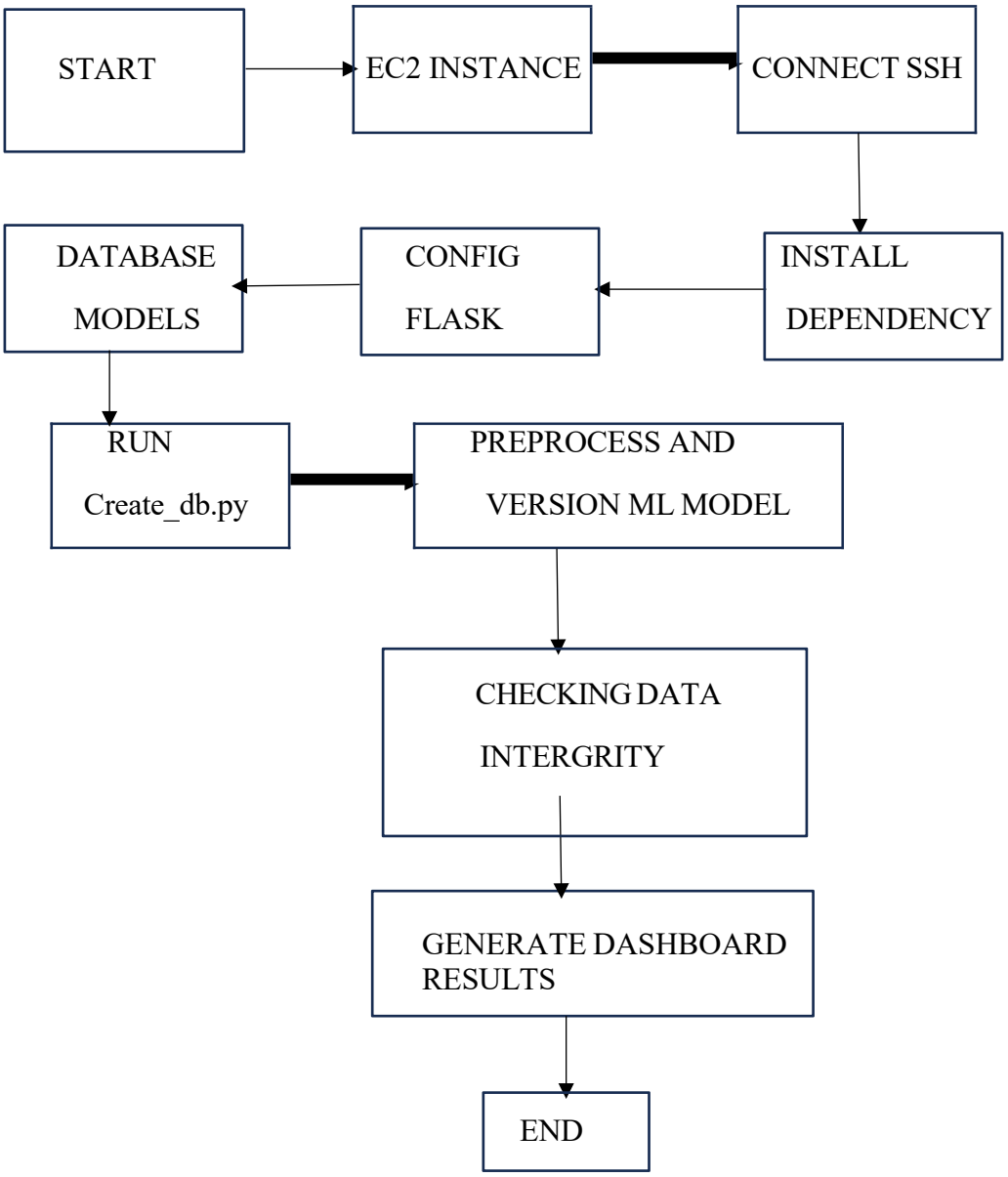


Figure 4.3 Activity Diagram for Data Engineering Module

depicts the workflow of the Data Engineering Module, starting with running the database file and preprocessing the dataset using the Flask-based ML model.

4.5 RESULTS OF DATA ENGINEERING

The Data Engineering module was successfully completed. A scalable and well-structured cloud data pipeline was implemented using AWS services for collecting, storing, and analyzing campus Wi-Fi usage data. Amazon S3 was used as the centralized storage layer, while AWS Lambda and Amazon Athena were integrated for processing and analytics.

Key outcomes:

- Reliable storage for Wi-Fi usage logs, device count records, bandwidth metrics, and access point load data.
- Proper folder structure and dataset organization in Amazon S3 for efficient retrieval and future scalability.
- Seamless integration between API Gateway, Lambda, S3, and Athena for real-time data operations.
- Historical data successfully prepared for anomaly detection and peak usage analysis.
- Data integrity maintained across stored records with secure access permissions and encryption support.

This module provides a strong foundation for the subsequent Cloud Engineering, AI Engineering, and Security Engineering modules. The data platform is now ready to support real-time monitoring, intelligent analytics, and anomaly detection features of the Campus Wi- Fi Usage Monitoring System.

CHAPTER - 5

MODULE 2 : CLOUD ENGINEERING

5.1 AIM

To deploy and configure a secure, scalable, and serverless cloud environment using AWS services for hosting the **Campus Wi-Fi Usage Monitoring and Anomaly Detection System**, ensuring reliable backend connectivity, secure data storage, efficient processing of Wi-Fi usage data, and smooth communication between frontend and cloud resources through proper configurations and access control mechanisms..

5.2 PROCEDURE FOR CLOUD ENGINEERING

Step 1: Log into the AWS Management Console and activate the AWS Academy Learner Lab environment.

Step 2: Create and configure cloud resources including Amazon S3 for data storage, AWS Lambda for backend processing, and Amazon API Gateway for secure API communication.

Step 3: Configure security settings such as IAM roles, bucket permissions, and API access controls to allow secure interaction between services.

Step 4: Set up the backend application environment and configure AWS Lambda functions to process incoming Wi-Fi metrics such as device count, bandwidth usage, and access point load.

Step 5: Install required dependencies and libraries for backend development, AWS SDK integration, and API communication.

Step 6: Establish secure communication between the application and cloud services using proper credentials and environment variables.

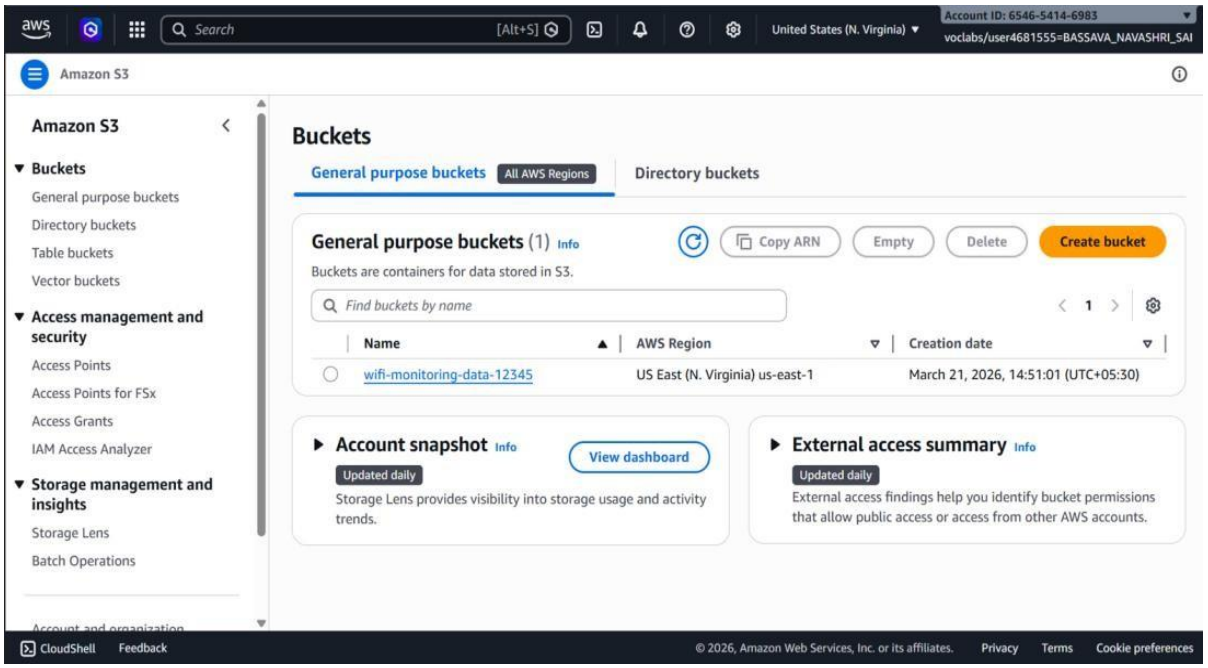
Step 7: Create and configure Amazon S3 bucket to store Wi-Fi logs, analytics outputs, historical records, and processed datasets.

Step 8: Implement access control using IAM roles and policies to ensure secure interaction between Lambda, S3, Athena, and monitoring services.

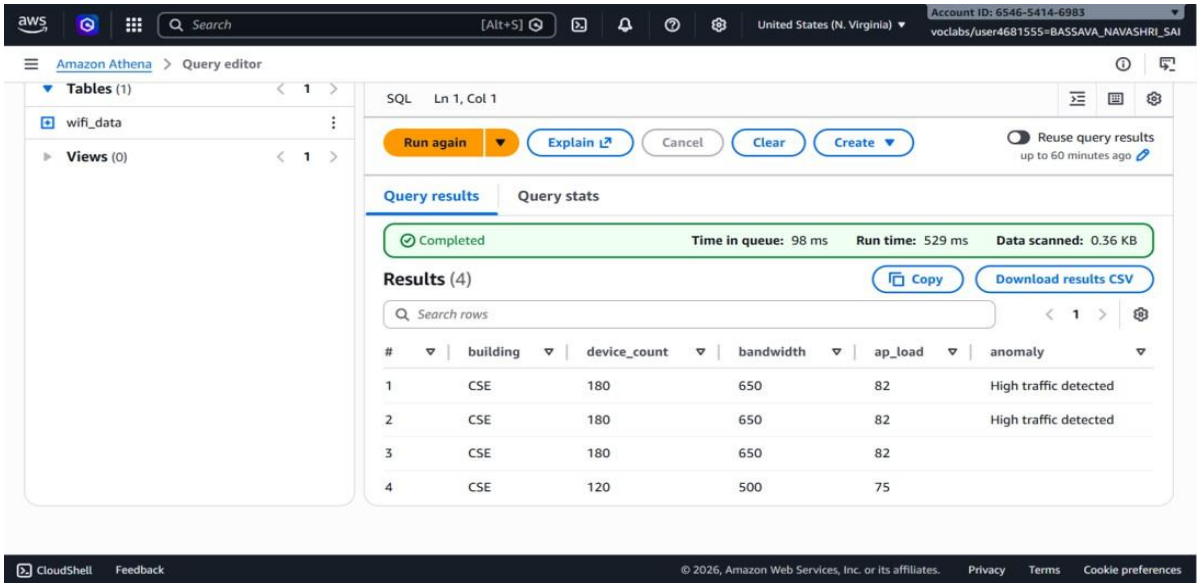
Step 9: Configure Amazon Athena to run SQL queries on stored Wi-Fi datasets .

5.3 SCREENSHOTS FOR CLOUD ENGINEERING

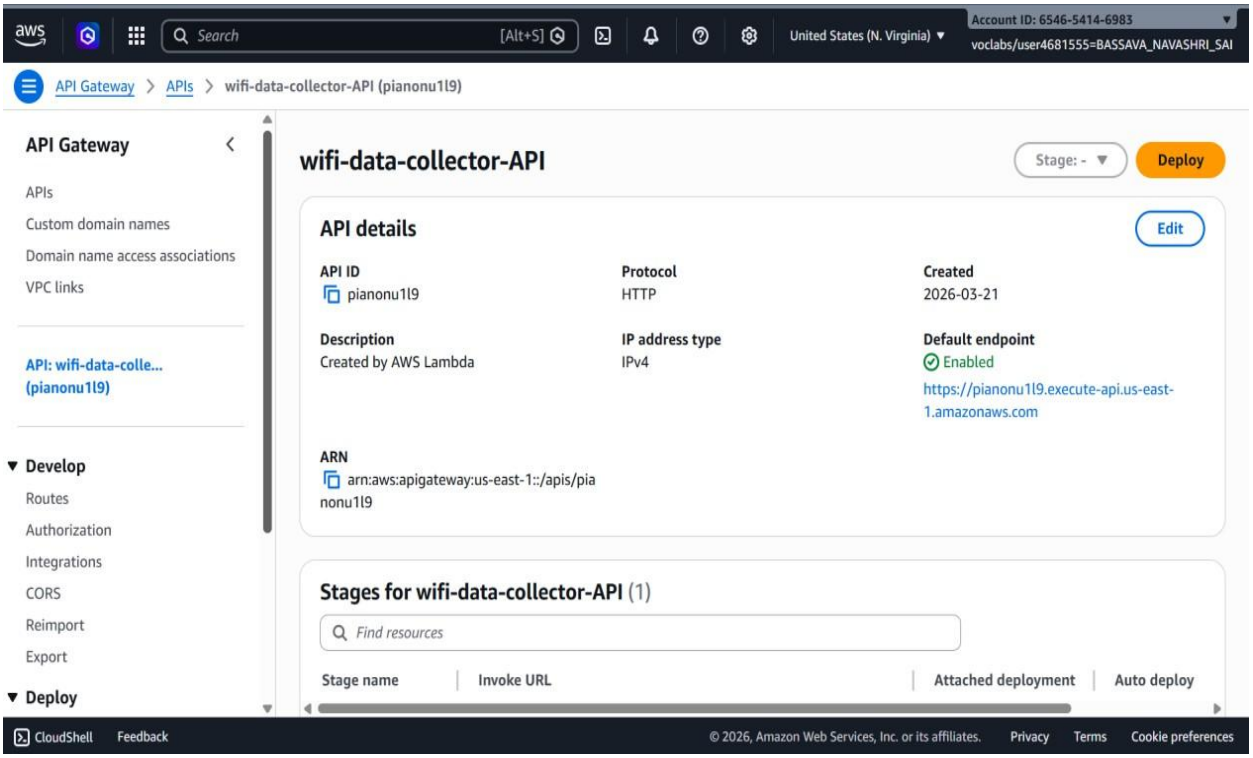
The following **4 screenshots** illustrate the important stages executed in this module:



Screenshot 5.1 : Amazon S3 Bucket Created for Centralized Storage of Wi-Fi Logs.



Screenshot 5.2 : Amazon Athena Query Results Showing Wi-Fi Usage and Anomaly Detection Output.



Screenshot 5.3 : Amazon API Gateway API Details Showing Endpoint URL and Configuration for Wi-Fi Data Collector API.

5.4 ACTIVITY DIAGRAM

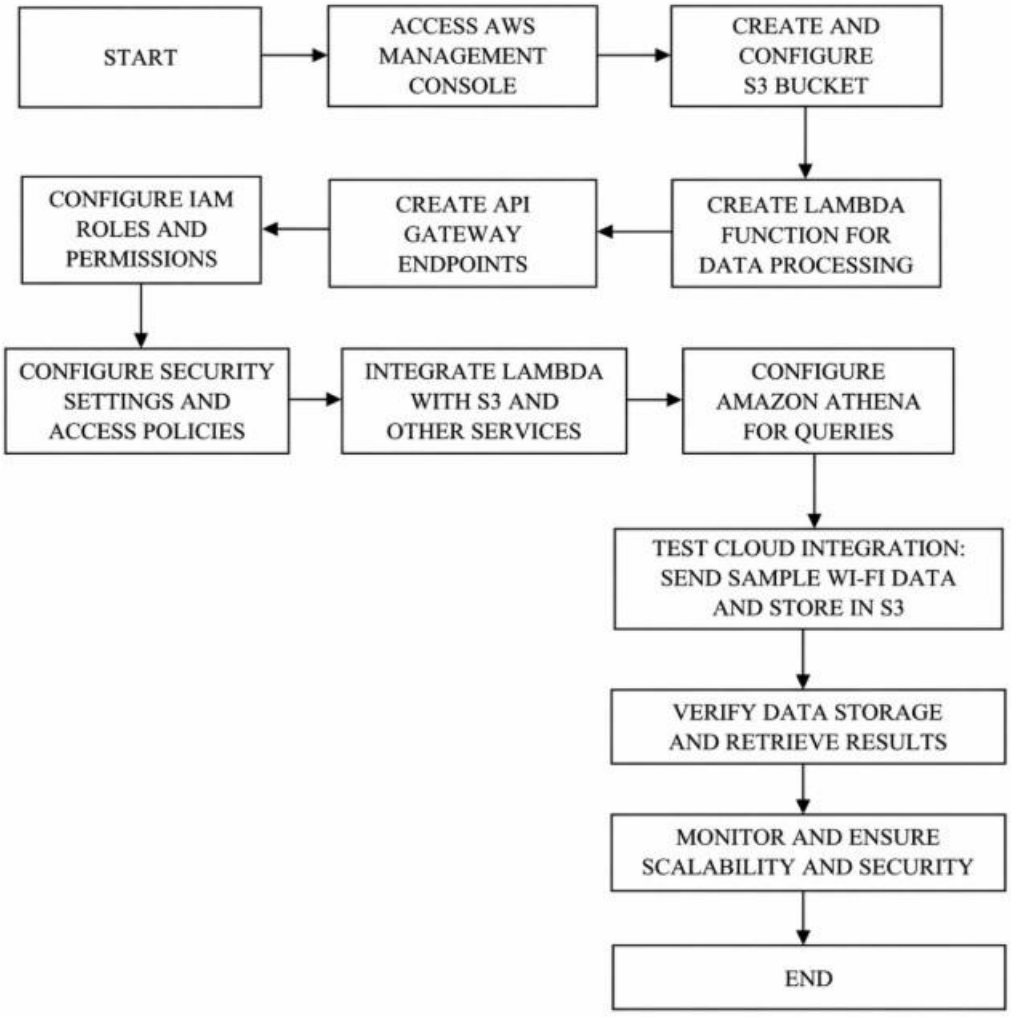


Figure 5.4 Activity Diagram for Cloud engineering

Fig. 5.5 depicts the activity flow of the Cloud Engineering module, beginning with accessing the AWS console and configuring core services such as Amazon S3, IAM, Lambda, API Gateway, and Athena.

5.5 RESULT OF CLOUD ENGINEERING

The Cloud Engineering module was successfully completed. The Campus Wi-Fi Usage Monitoring and Anomaly Detection System is now effectively integrated with AWS cloud services and functioning reliably.

Key Achievements:

- A secure and scalable serverless cloud infrastructure was established using AWS Lambda, API Gateway, and Amazon S3.
- The backend application was successfully connected to cloud services, enabling real- time collection, storage, and retrieval of Wi-Fi usage data.
- Cloud storage was configured using Amazon S3 to handle network logs, historical records, analytics outputs, and monitoring datasets.
- Secure communication and access control mechanisms were implemented using HTTPS configurations and IAM roles/policies.
- The system demonstrated stable performance with successful API responses, Lambda execution, and reliable data storage operations.
- Amazon Athena was integrated successfully to perform SQL-based analytics on stored Wi-Fi data.
- The cloud setup is now ready to support data engineering pipelines, anomaly detection models, and intelligent query features in subsequent modules.

CHAPTER 6

MODULE 3 : AI ENGINEERING

6.1 AIM

To design and integrate intelligent analytics models for detecting abnormal Wi-Fi usage patterns in the Campus Wi-Fi Usage Monitoring and Anomaly Detection System by analyzing network metrics such as device count, bandwidth usage, access point load, and traffic trends, ensuring accurate anomaly detection, peak hour prediction, minimal false alerts, and efficient real-time performance.

6.2 PROCEDURE FOR AI ENGINEERING

Step 1: Set up the development environment and install required ML libraries such as scikit- learn, numpy, pandas, and joblib.

Step 2: Collect and preprocess Wi-Fi usage data including device count, bandwidth metrics, access point load, timestamps, and building-wise traffic records.

Step 3: Perform feature engineering by converting raw logs into meaningful inputs such as event frequency, duration, and anomaly counts.

Step 4: Train a machine learning model (e.g., Decision Tree, Random Forest, or threshold-based classifier) to classify network behavior as Normal, High Load, or Anomaly.

Step 5: Evaluate the model using test data to ensure accuracy and reduce false positives in anomaly detection.

Step 6: Save the trained model as a serialized (.pkl) file for efficient loading during runtime.

Step 7: Load the trained model into the backend application and integrate it with Wi-fi data processing logic.

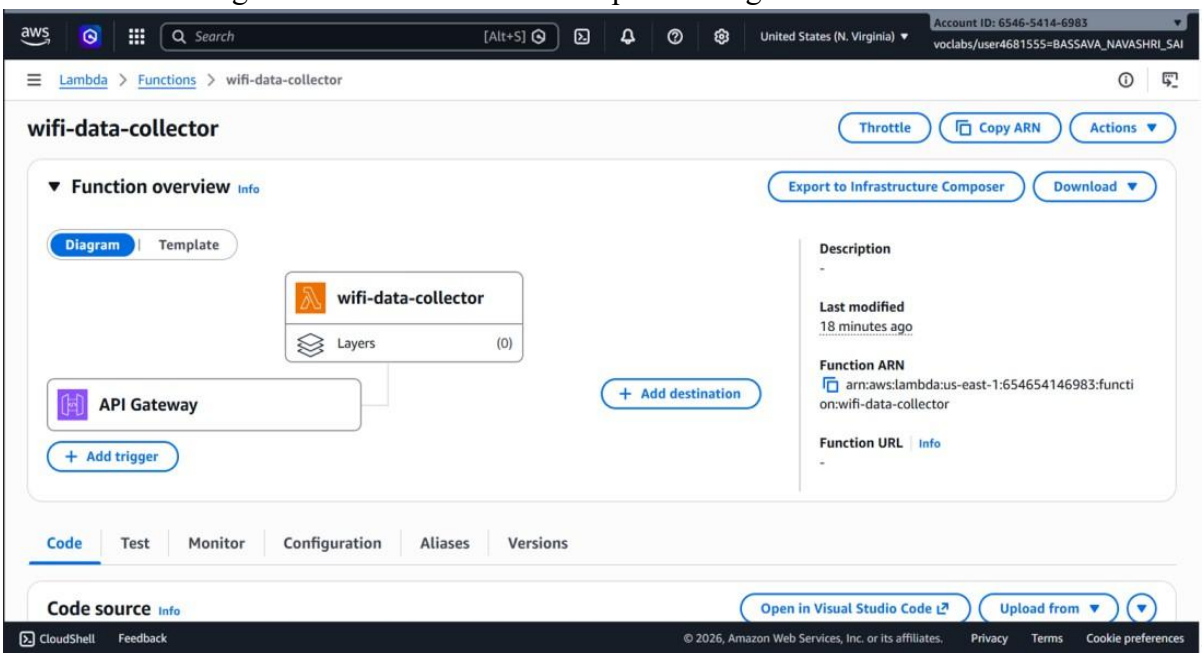
Step 8: Trigger model inference in real time by feeding current Wi-Fi metrics to generate network status predictions and anomaly alerts.

Step 9: Store model outputs along with Wi-fi logs for further analysis and reporting.

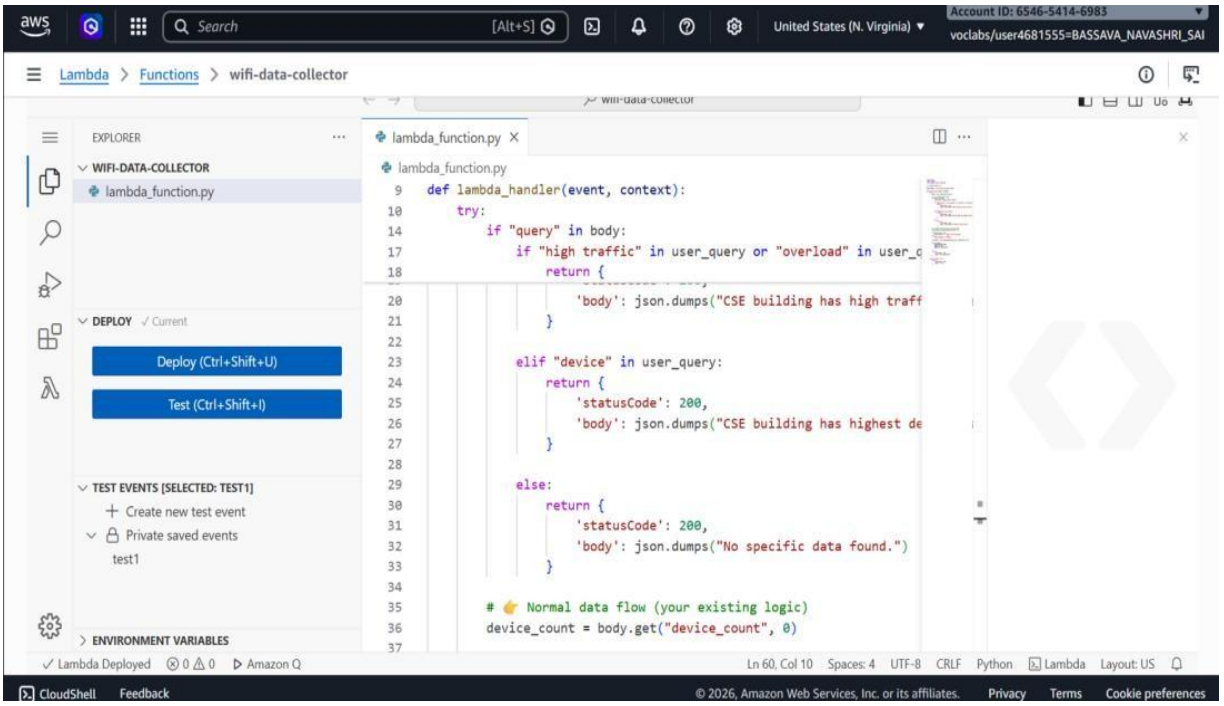
Step 10: Validate system performance by testing real-time predictions, peak hour forecasting, low latency responses, and reliable anomaly detection.

6.3 SCREENSHOTS OF AI ENGINEERING

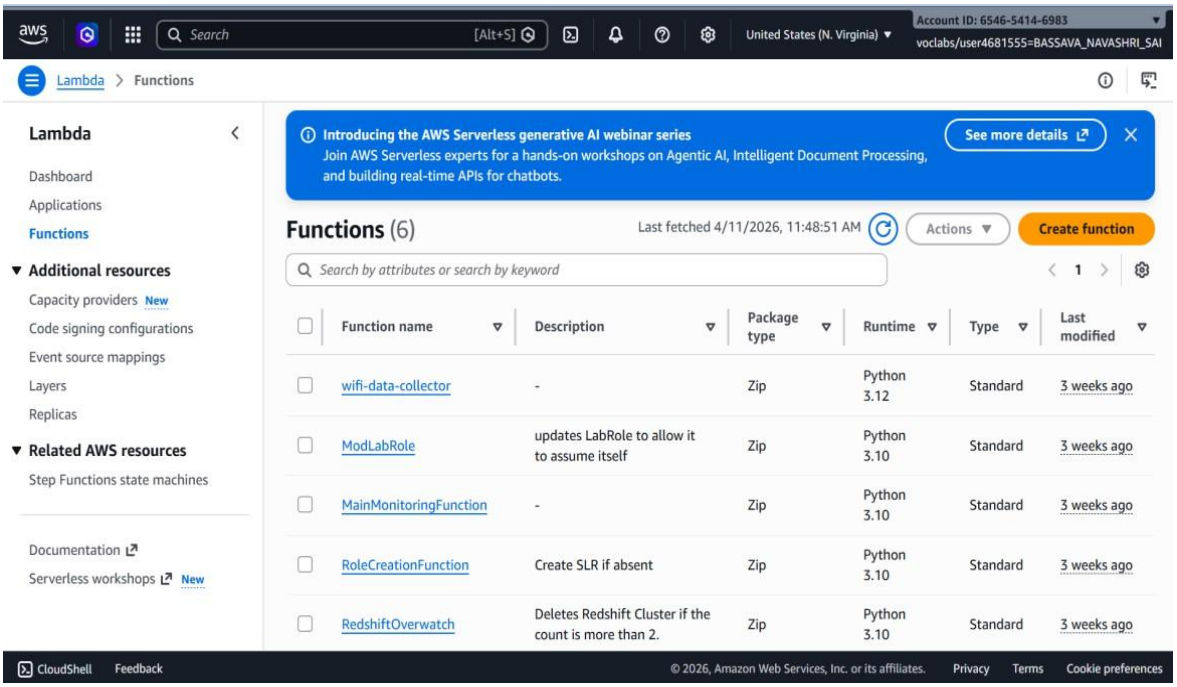
The following screenshots illustrate the important stages executed in this module:



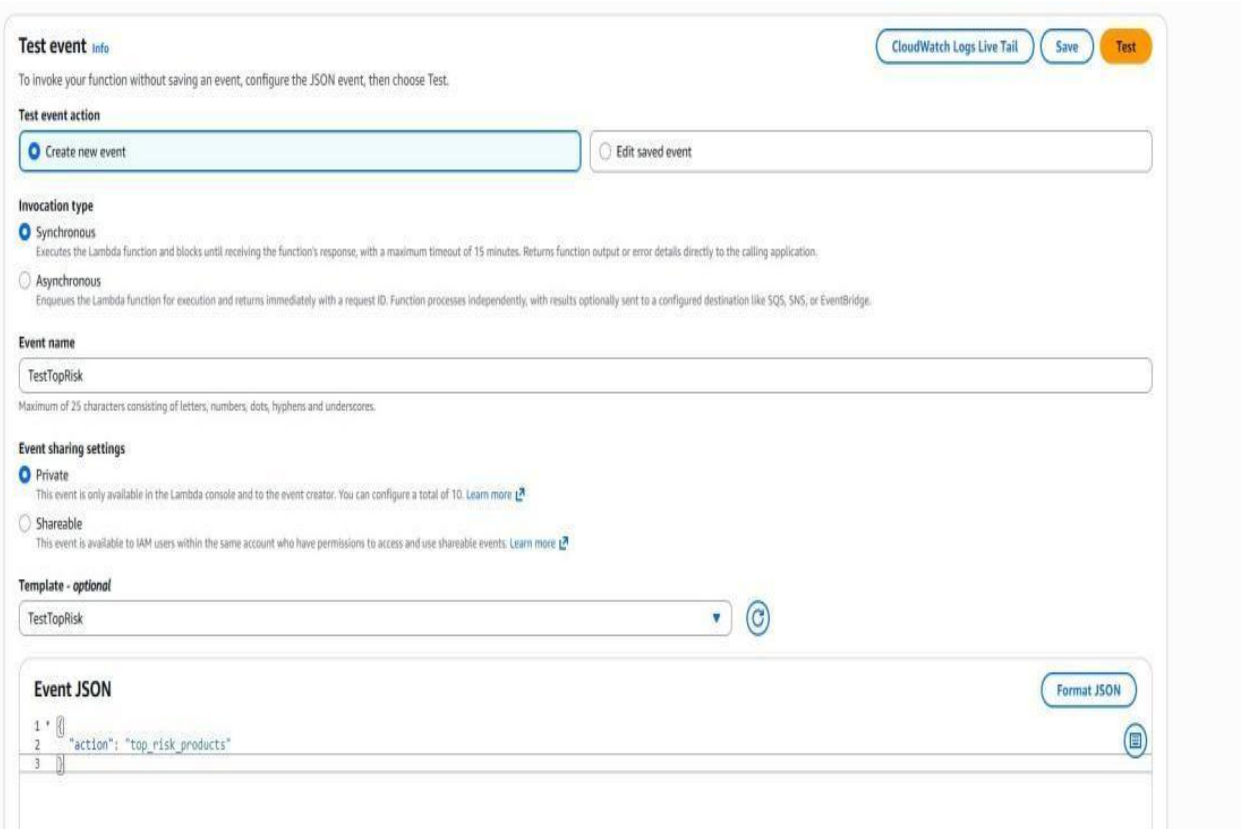
Screenshot 6.1 : AWS Lambda Deployment of Wi-fi Data Collector Integrated with API Gateway..



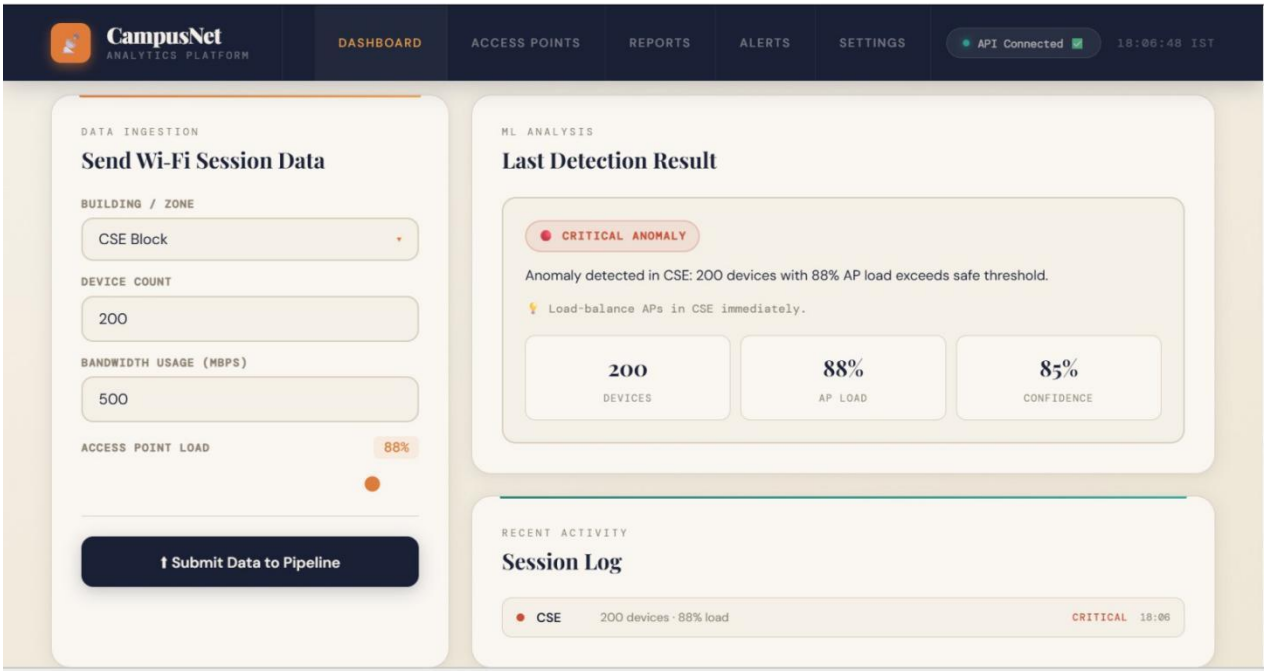
Screenshot 6.2 : Lambda Function Source Code for Wi-Fi Data Processing and Query Handling



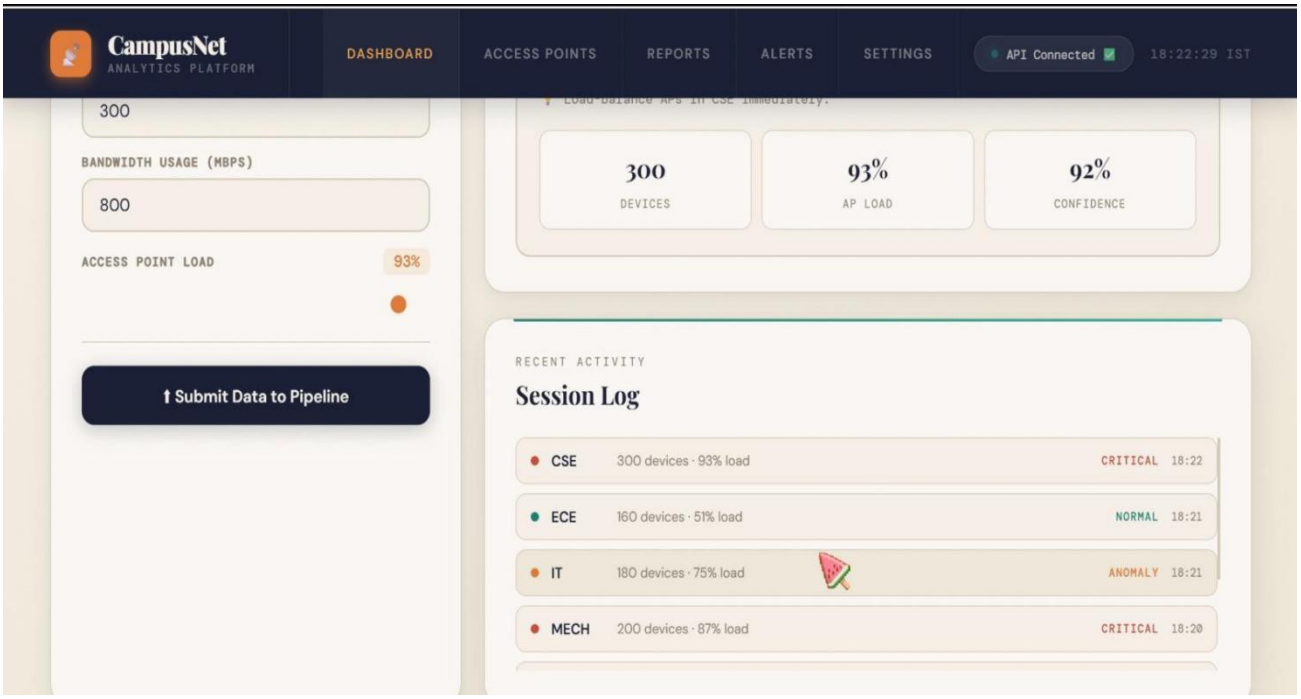
Screenshot 6.3 : AWS Lambda Functions Dashboard with Deployed Wi-Fi Data Collector Function



Screenshot 6.4: AWS Lambda Test Event Configuration with Sample JSON Input.



Screenshot 6.5 : Critical anomaly scenario indicating excessive device density and severe network congestion.



Screenshot 6.6 : Session log displaying real-time network status across multiple locations with classification of normal, anomalous, and critical conditions.

6.4 ACTIVITY DIAGRAM

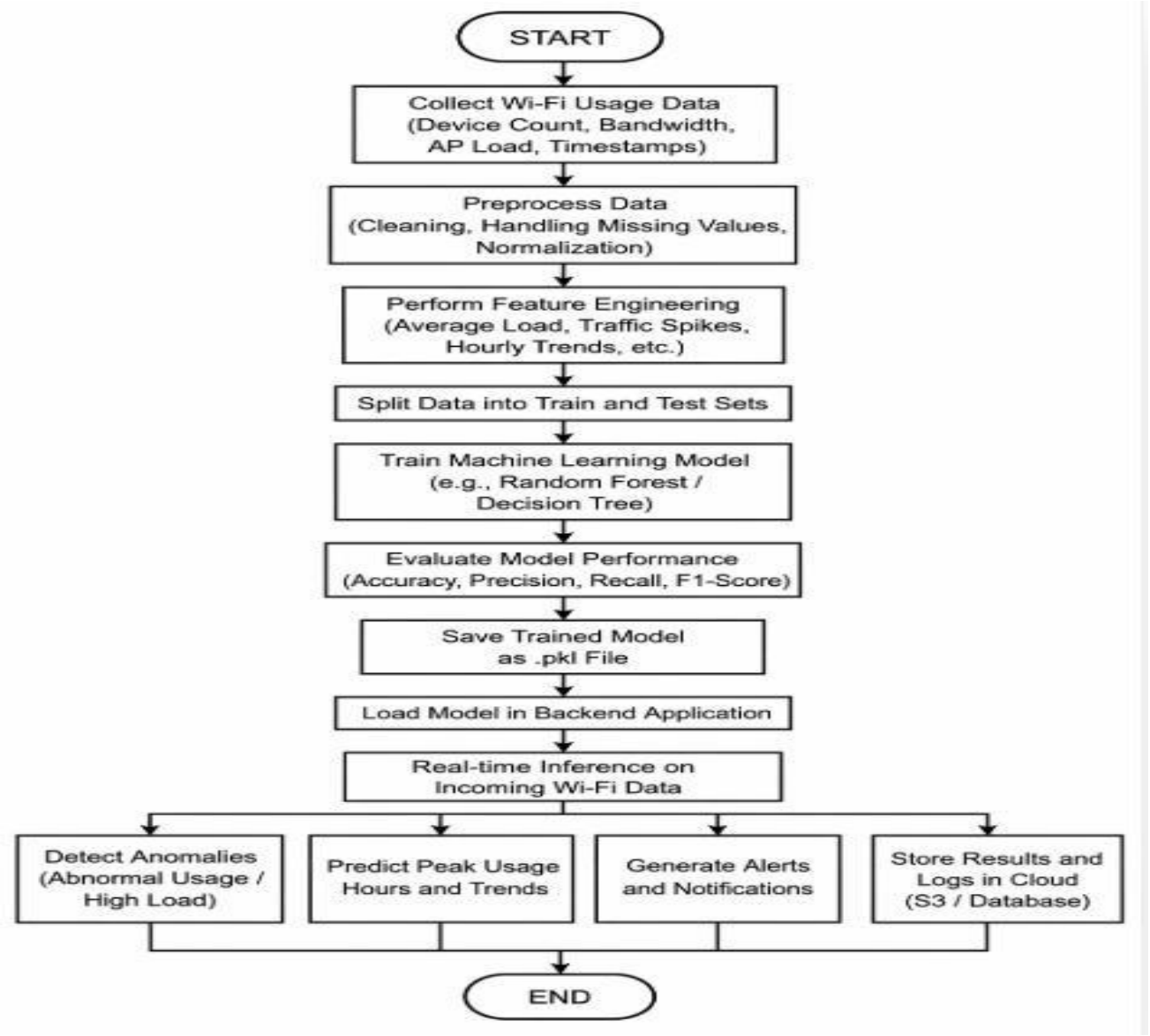


Figure 6.7 : Activity Diagram for AI Engineering Module

Fig. 6.7 depicts the activity flow of the AI Engineering Module, starting with collecting Wi-Fi usage data and preprocessing it through cleaning, normalization, and feature engineering.

6.5 RESULT OF AI ENGINEERING MODULE

The AI Engineering module was successfully completed. The machine learning model for detecting abnormal Wi-Fi usage patterns was trained, integrated, and deployed into the Campus Wi-Fi Usage Monitoring and Anomaly Detection System.

Key Achievements:

- Machine learning model (Random Forest / Decision Tree / Threshold Classifier) successfully implemented for detecting network anomalies based on Wi-Fi usage metrics.
- Feature-based analysis enabled using inputs such as device count, bandwidth usage, access point load, traffic spikes, and session trends.
- The trained model was serialized and integrated into the backend for efficient loading and real-time inference.
- The system now generates network status predictions such as Normal Load, High Load, or Anomaly for each monitored access point.
- Model inference is performed with low latency, ensuring real-time monitoring without affecting overall system performance.
- AI-based detection improves accuracy compared to manual monitoring methods and reduces false alerts.
- The module also supports peak usage prediction and proactive network management for better campus connectivity.

CHAPTER 7

MODULE 4: SECURITY ENGINEERING

7.1 AIM

To implement a secure and reliable multi-layered security framework for the Campus Wi-Fi Usage Monitoring and Anomaly Detection System using authentication, encryption, secure APIs, IAM access control, and AWS security services to protect network usage data, analytics reports, cloud resources, and system operations from unauthorized access and cyber threats.

7.2 PROCEDURE FOR SECURITY ENGINEERING

Step 1: Configure secure user authentication for the admin dashboard using session management or token-based authentication.

Step 2: Implement role-based access control to restrict system access for administrators, network staff, and authorized users only.

Step 3: Secure all API communications using HTTPS to ensure encryption of data in transit and use REST API

Step 4: Configure Amazon S3 with secure bucket permissions and restricted public access policies.onfigure

Step 5: Enable encryption for data stored in Amazon S3 to protect Wi-Fi logs, analytics reports, and historical records.

Step 6: Implement input validation and sanitization to prevent common web vulnerabilities such as SQL Injection, XSS, and invalid API requests.

Step 7: Apply secure headers and request validation mechanisms within the backend application for security purpose

Step 8: Configure AWS IAM roles and policies to control access between Lambda, API Gateway, S3, Athena, and CloudWatch.

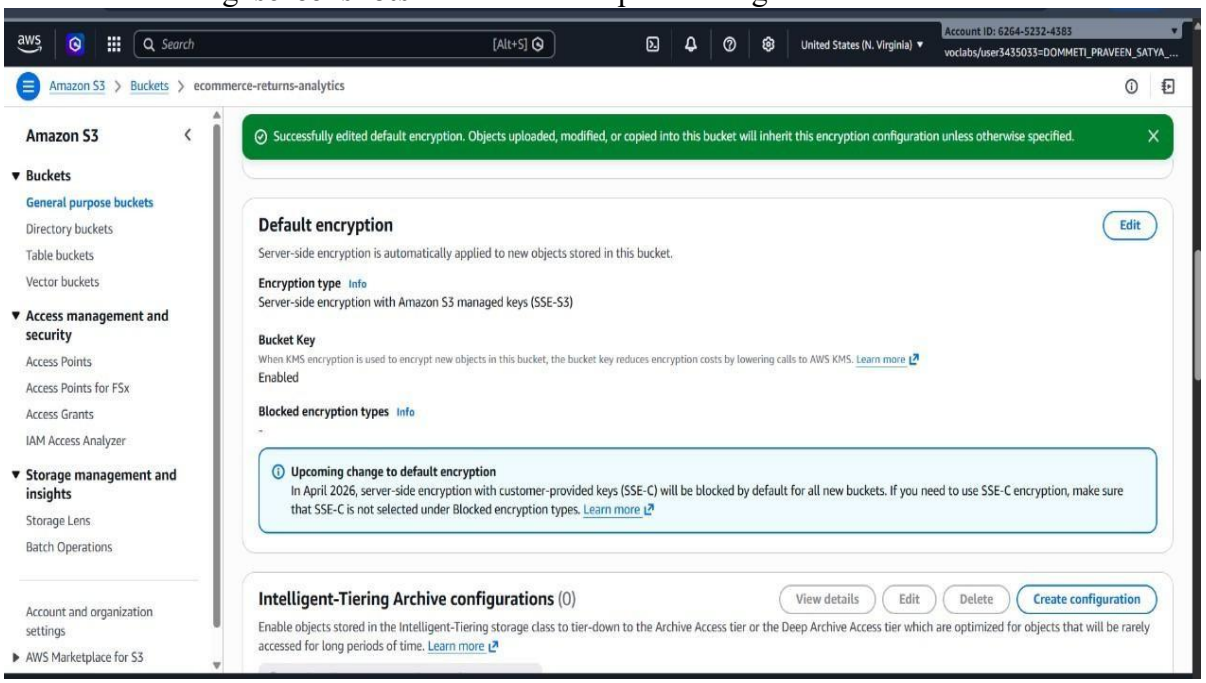
Step 9: Restrict network access using proper security configurations to allow only required inbound and outbound communication.

Step 10: Implement secure session handling and automatic timeout mechanisms for dashboard users to prevent unauthorized access

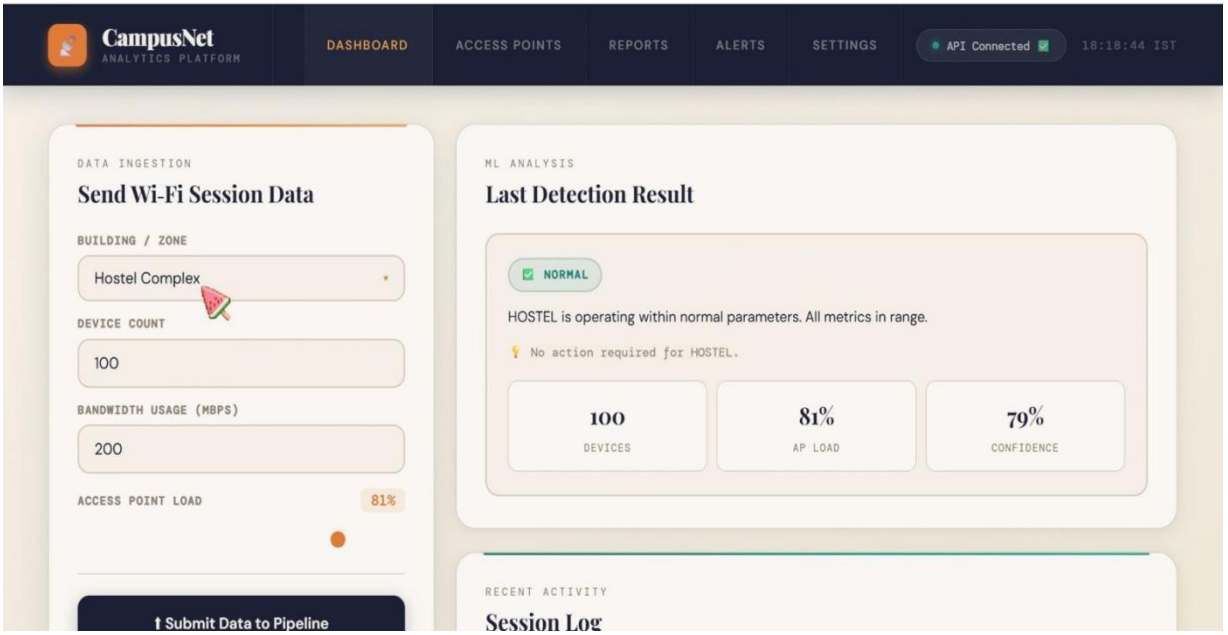
Step 11: Log security-related events and monitor suspicious activities using AWS CloudWatch.

7.3 SCREENSHOTS OF SECURITY ENGINEERING

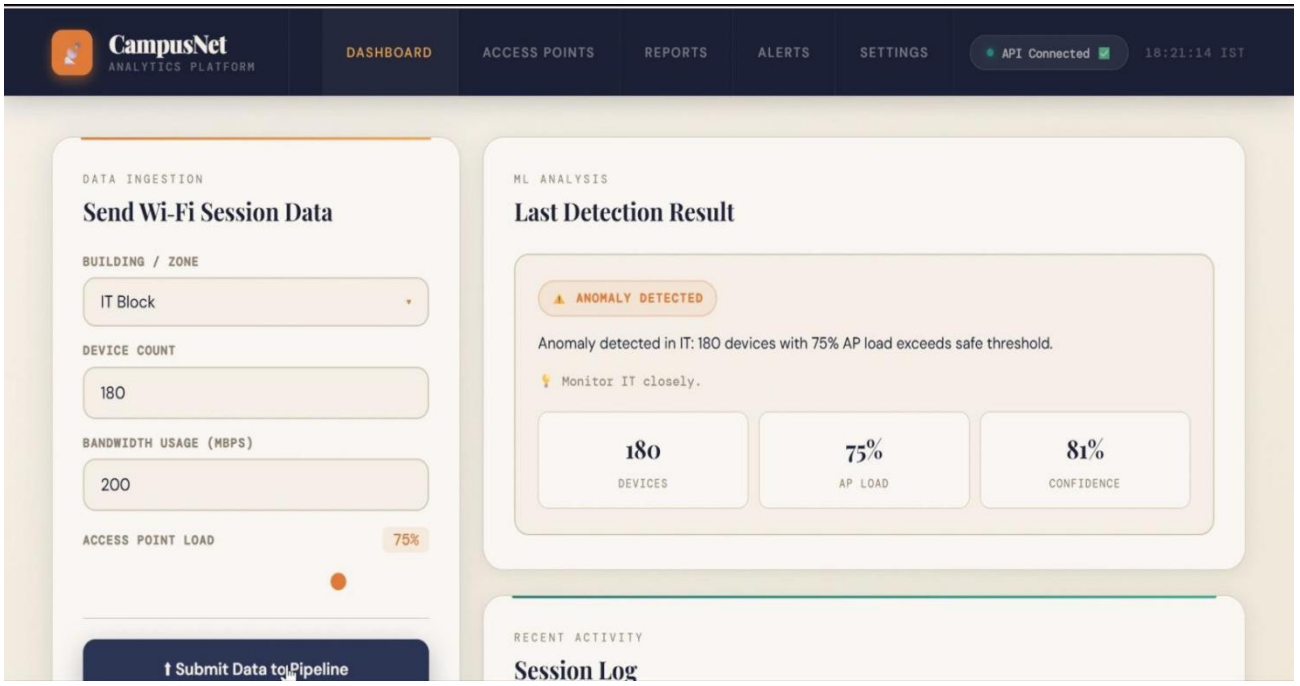
The following **screenshots** illustrate the important stages executed in this module:



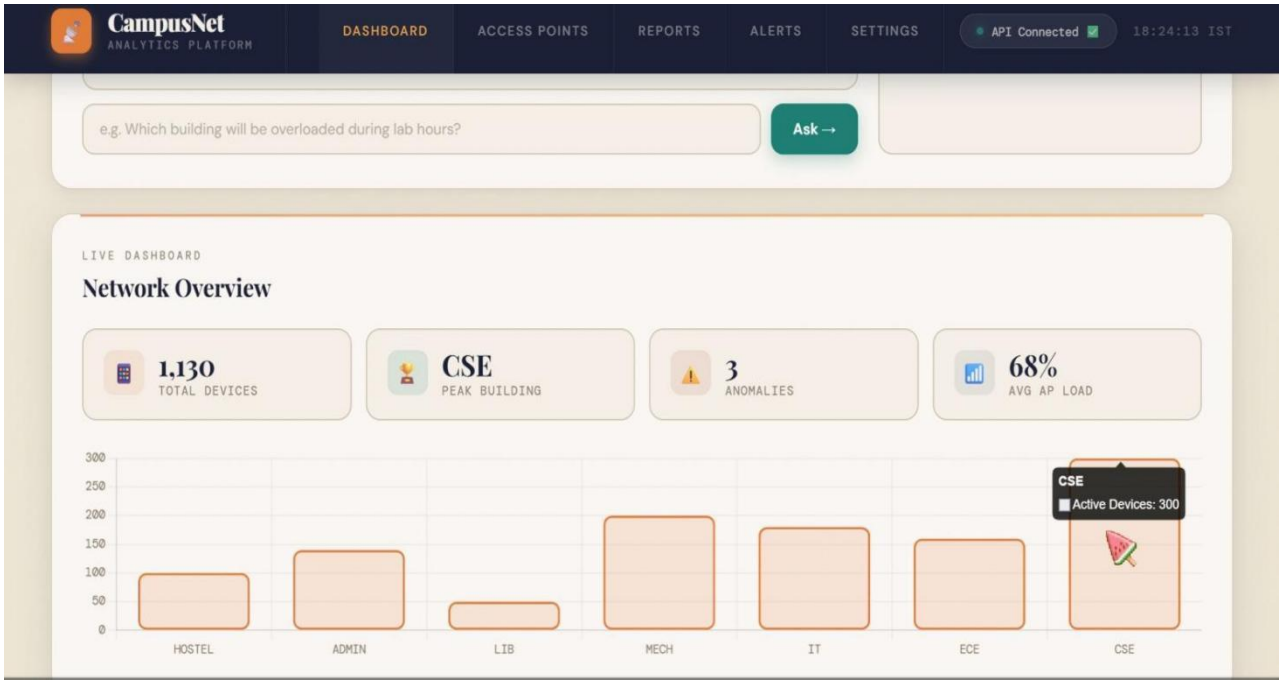
Screenshot 7.1 : Amazon S3 Security Configuration with Default Encryption Enabled.



Screenshot 7.2 : Dashboard illustrating normal network operation with all parameters within acceptable limit



Screenshot 7.3 : Visualization of anomaly detection triggered by increased device count and access point load.



Screenshot 7.4 : Live dashboard presenting overall network metrics, including total devices, detected anomalies, average load, and building-wise usage distribution.

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7.4 ACTIVITY DIAGRAM

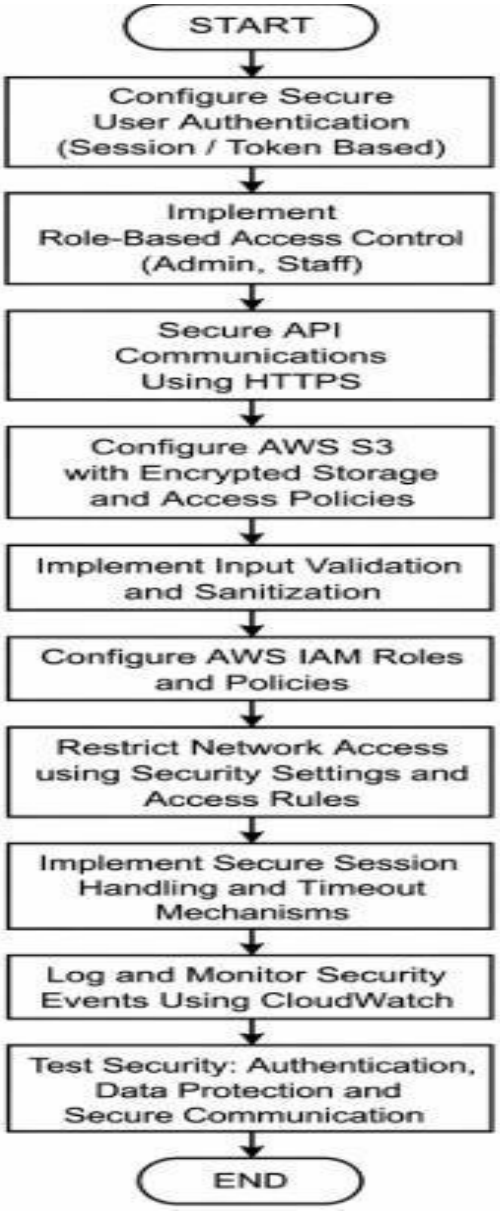

Figure 7.5 : Activity Diagram for Security Engineering Module

This fig. 7.5 represents an activity diagram for a campus monitoring and anomaly detection system. It shows the workflow starting from secure user authentication and role-based access for admins and staff.

7.5 RESULT OF SECURITY ENGINEERING

The Security Engineering module was successfully completed. Robust security mechanisms were implemented across the Campus Wi-Fi Usage Monitoring and Anomaly Detection System to ensure secure data handling, protected cloud resources, and reliable network analytics operations.

Key Achievements:

- Secure authentication mechanisms implemented using session management and token-based access control for authorized dashboard users.
- Role-based access control enforced to differentiate permissions between administrators, network staff, and authorized users.
- Data encryption applied for secure communication and storage, protecting sensitive information such as Wi-Fi logs, analytics reports, and monitoring data.
- AWS security configurations implemented using Amazon S3, AWS Lambda, IAM roles, and API Gateway with restricted access policies.
- Network-level security ensured through proper configuration of secure API access, HTTPS communication, and controlled access rules.
- Protection mechanisms against common web vulnerabilities such as SQL Injection, XSS, CSRF, and invalid requests were successfully implemented.
- CloudWatch logging and monitoring enabled to track suspicious activities, audit events, and maintain system reliability.

CONCLUSION

The development of the Campus Wi-Fi Usage Monitoring & Anomaly Detection System demonstrates an effective approach to managing and analyzing network usage in modern campus environments. The project successfully integrates data engineering, cloud computing, and machine earning concepts to create a scalable and intelligent monitoring system.

By implementing a structured data pipeline, the system ensures efficient collection, storage, and processing of Wi-Fi usage data. The use of cloud services such as Amazon S3, AWS Lambda, API Gateway, and Amazon Athena enables scalable and reliable system performance. This cloud-based architecture allows the system to handle large volumes of data while maintaining efficiency and flexibility [11].

The integration of anomaly detection techniques helps in identifying unusual patterns such as high network load and abnormal device activity. This improves the ability to monitor network behavior and supports proactive decision-making. Additionally, the query-based assistant enhances user interaction by providing meaningful insights into Wi-Fi usage data.
Security measures such as IAM-based access control, data encryption, and monitoring using CloudWatch ensure safe and reliable system operation. These features enhance data protection and system transparency.

In conclusion, the system provides a practical and scalable solution for monitoring campus Wi- Fi networks. It highlights the importance of integrating cloud technologies and intelligent analysis to build efficient and reliable systems. This project can be further extended with advanced machine learning models and real-time analytics to improve accuracy and performance in future implementations.

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